

# ECE 447

## Lesson 37

### Binary System

### Performance, Part 2



UNITED STATES  
AIR FORCE  
ACADEMY

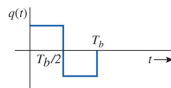
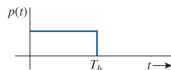
# SCHEDULE AND ADMIN

- **Schedule.**
  - Lesson 37 - Binary digital system performance (Part 2)
  - Lesson 38 - Error correction
  - Lesson 39 - SDR Lab 8: Capstone image transfer
  - Lesson 40 - Advanced topics: OFDM, MIMO, CDMA
  - Lesson 41 - Course review
- **Admin**
  - **Lab 7.** Due tonight (Lesson 37, 23 Nov) to Gradescope.
  - **HW8.** Due Lesson 39 (Thu, 3 Dec) to Gradescope.

# REVIEW

## Matched Filters and Binary Signaling Performance

- Why?
- General binary case for real-valued pulses:  $h(t) = p(T_b - t) - q(T_b - t)$
- $P_e = P_b = Q\left(\sqrt{\frac{E_p + E_q - 2E_{pq}}{2N_0}}\right)$ , where  $E_{pq} = \int_0^{T_b} p(t)q(t)dt$  is the inner product or correlation of the two pulses
- Example problem: Find the matched filter impulse response  $h(t)$  and calculate the total bit error probability  $P_e$  assuming equiprobable transmission of 0 or 1



# BINARY SIGNALING PERFORMANCE (AWGN)

## Polar signaling, $q(t) = -p(t)$

- Use general equations for  $P_e$  and  $a_0$

- $E_p = E_q = \int_0^{T_b} |p(t)|^2 dt$

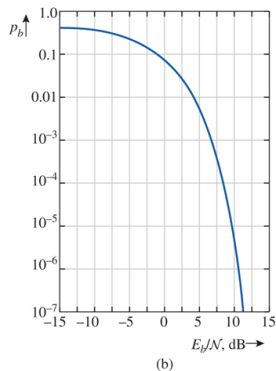
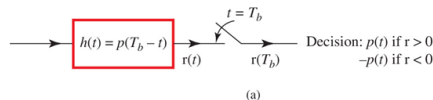
- $E_{pq} =$

- $P_e =$

- $a_0 =$

- $E_b = E_p \cdot P(m = 1) + E_q \cdot P(m = 0) =$

- BER depends on pulse energy not shape!



# BINARY SIGNALING PERFORMANCE (AWGN)

**Orthogonal signaling**,  $q(t)$  and  $p(t)$  are orthogonal over  $(0, T_b)$

- Use general equations for  $P_e$  and  $a_0$

- $E_q =$

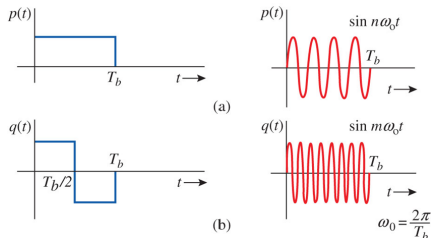
- $E_{pq} =$

- $P_e =$

- $E_b = \frac{E_p + E_q}{2}$  assuming equiprobable

- $a_0 = \frac{1}{2}(E_p - E_q)$  (general form)

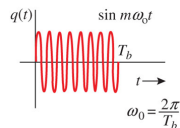
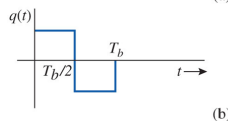
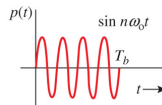
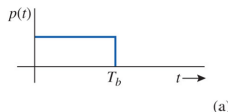
- Requires twice as much energy per bit (3db more power) to achieve same performance as polar signaling!



# BINARY SIGNALING PERFORMANCE (AWGN)

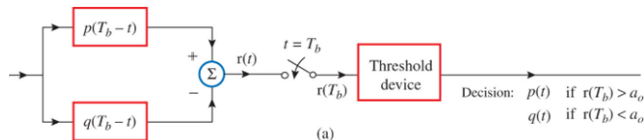
## On-Off Signaling, special case of orthogonal signaling

- Use general equations for  $P_e$  and  $a_0$
- $E_p = E_q$ ? Don't know, just leave  $E_p$  and  $E_q$
- $E_{pq} =$
- $P_e =$
- $E_b = \frac{E_p + E_q}{2} =$
- $a_0 = \frac{1}{2}(E_p - E_q) =$



# RECEIVERS

## Optimum binary threshold detector (baseband)



## Coherent detector for bandpass signals

